These are just a few examples of misconceptions about Elasticsearch. As mentioned earlier, you must carefully evaluate your specific requirements and use case before choosing Elasticsearch or any other technology.

1.5 Popular adoption

A long list of organizations use Elasticsearch for everything from searching to business analysis, log analytics, security alert monitoring, and application management, as well as using it as a document store. Let's consider some of these organizations and how they put Elasticsearch to use in their operations.

Uber powers its rider and event prediction using Elasticsearch. It does so by storing millions of events, searching through them, and analyzing the data at a near-real-time rate. Uber predicts demand based on location, time, day, and other variables, including taking past data into consideration. This helps Uber deliver rides pretty.

Netflix adopted the Elastic Stack to provide customer insights to its internal teams. It also uses Elasticsearch for log event analysis to support debugging, alerting, and managing its internal services. Email campaigns and customer operations are all backed by the Elasticsearch engine. The next time you receive an email from Netflix mentioning a newly added movie or TV series, keep in mind that the campaign analytics behind that simple email were all supported by Elasticsearch.

PayPal embraced Elasticsearch as a search engine to allow customers to store and search through their transactions. The company has implemented transaction search features along with analytics for use by merchants, end customers, and developers.

Similarly, the online e-commerce company eBay adopted Elasticsearch to support full-text searching by end users. As users, we are using Elasticsearch directly when searching through eBay's inventory. The company also uses the Elastic Stack for analytics, log monitoring, and storing transactional data in a document store.

GitHub, a popular code repository for developers, indexes its 8 million (and counting) code repositories—consisting of over 2 billion documents—with Elasticsearch to enable a powerful search experience for its users. Similarly, Stack Overflow uses Elasticsearch to provide developers with quick and relevant answers, and Medium (a popular blog platform) uses the Elastic Stack to serve reader queries in a near-real-time mode.

Before we wind up this chapter, it's only fair to touch on a recent trending topic: generative artificial intelligence (AI) tools such as OpenAI's ChatGPT and Google's Bard. The introduction of these tools will change the search space dramatically, in my opinion. Let's discuss their effects on modern search, including search engines like Elasticsearch.

1.6 Generative AI and modern search

Unless you are living in a cave, you have undoubtedly heard about a recent internet revolution: ChatGPT. ChatGPT is a generative AI tool that was developed and released by the OpenAI team in November 2022. In my 25 years of IT experience, I've never seen a tech tool light up the internet like ChatGPT. Is isn't often that a technically